

REAL-TIME FACE TRACKING

Using Raspberry Pi 4 Versus Raspberry Pi 5 With Hailo-8L

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Real-time face tracking has evolved from a research novelty into a practical tool for interactive robotics, surveillance, and automation. Advances in single-board computers and AI accelerators have enabled compact systems to detect and track human faces with

remarkable speed and accuracy. This face-tracking system is built on the Raspberry Pi 5, with optional Hailo-8L acceleration, and compares its performance against Raspberry Pi 4, demonstrating how hardware choices affect tracking responsiveness and precision.

Face tracking in this system is not designed to identify or recognise a specific individual. Its purpose is to detect human faces within the camera's field of view and dynamically track their movement. The objective of the system is to move the pan-tilt-mounted camera to locate human faces, lock onto the region with the highest face

density, and count the detected faces. If no faces are present, the system criss-crosses the visual frame to locate them, producing an interactive, almost lifelike behaviour reminiscent of R2-D2 in *The Empire Strikes Back* movie.

Equipped with a quad-core CPU and up to 8GB or 16GB RAM, it can handle computer vision workloads using OpenCV, TensorFlow Lite, or lightweight YOLO models. For applications that demand higher speed and reliability, a Hailo-8L AI accelerator can be added, providing an additional 8 or 16 TOPS of AI performance. In comparison, microcontroller platforms such as Arduino, ESP32, or MaixCam cannot achieve this level of performance.

Fig. 1 demonstrates real-time face detection in a crowded scene, where multiple human faces are simultaneously identified and highlighted with bounding boxes. Fig. 2 shows the author's prototype. The components required to build the system are listed in the Bill of Materials table.

Cooling requirements

A forced-fan cooler is normally required for Raspberry Pi operation and is essential for the Raspberry Pi 5. During intensive AI workloads, both the Pi 5 and the Hailo-8L AI module generate significant heat. As the Hailo-8L module is mounted openly above the Pi 5 processor, only a specific type of fan cooler is compatible with the Pi 5, while the AI module itself may not require separate active cooling. However, for additional thermal safety and



Fig. 1: Multiple human faces highlighted with bounding boxes



Fig. 2: Author's prototype